Programming Fundamentals

Lab 2 Submission

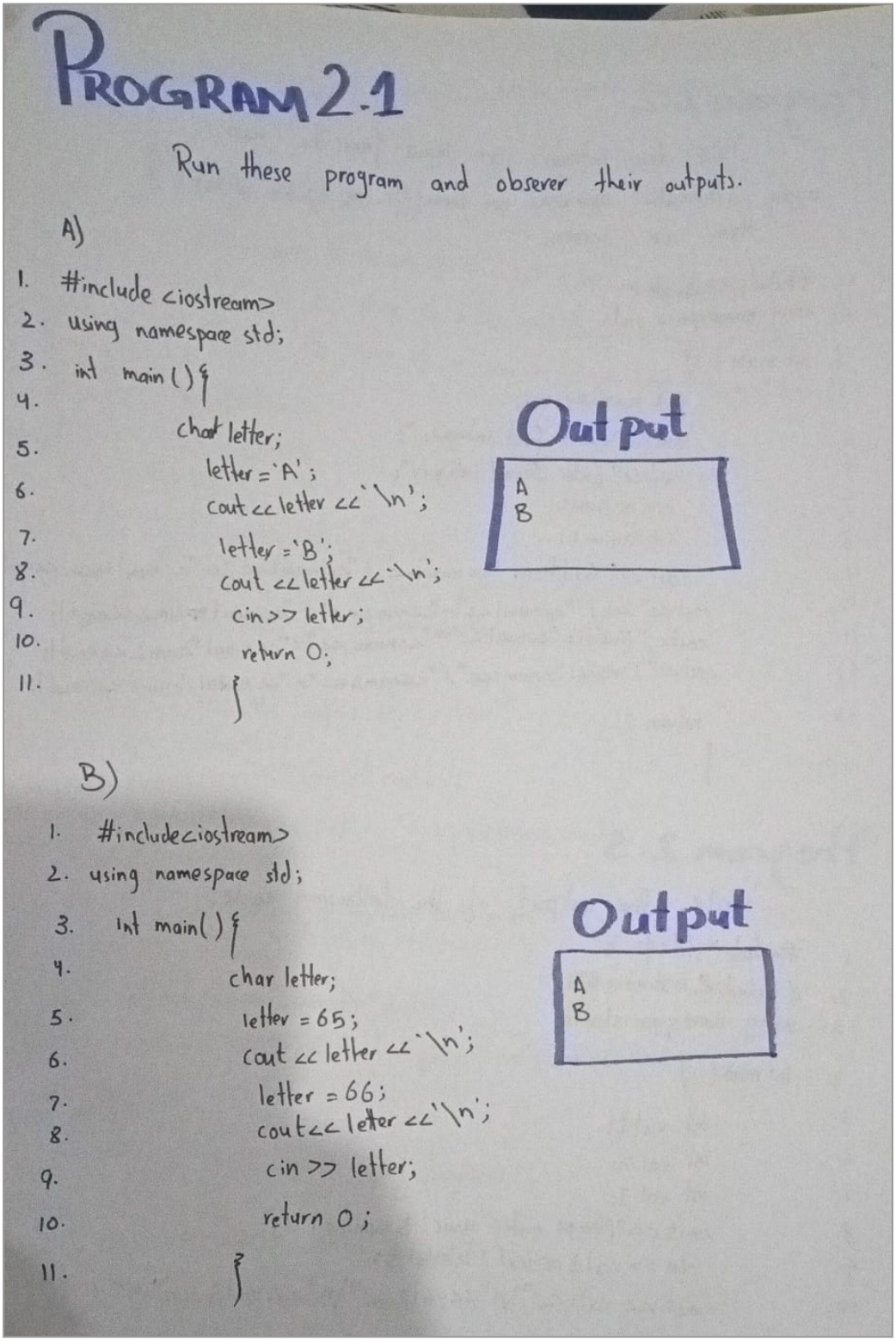
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Date	11 October 2026

Program 2.1

Question

Type and save the following programs in Visual Studio. Run these programs and observe their output.

Handwritten Code



Code Written on Phone / Desktop (Program A)

```cpp
#include <iostream>
using namespace std;
int main() {
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Output (Program A)

```
A
B
```

Code Written on Phone / Desktop (Program B)

```cpp
#include <iostream>
using namespace std;
int main() {
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Output (Program B)

```
A
B
```

Explanation

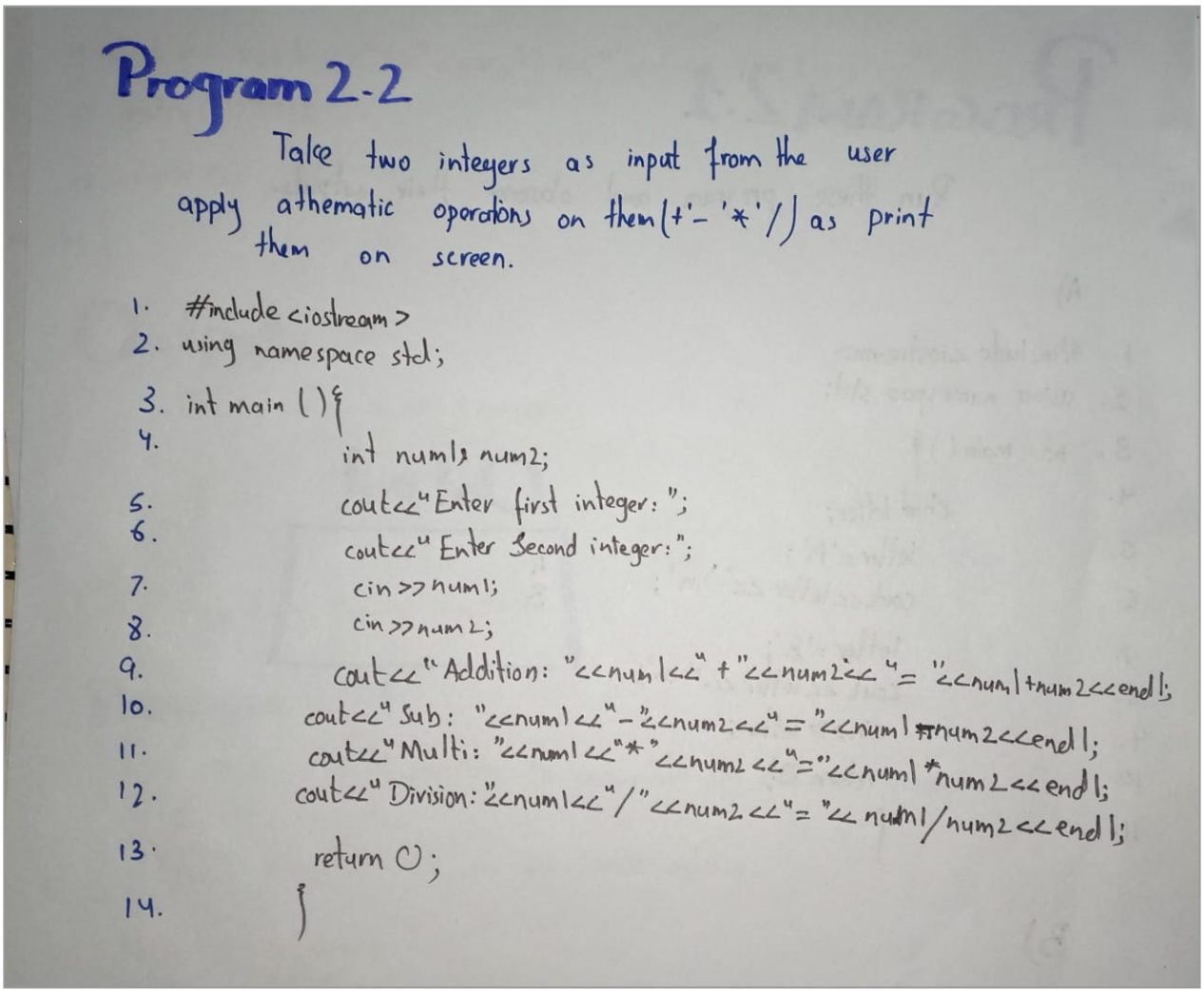
- Program A stores the characters `'A'` and `'B'` in a `char` variable and prints them, so the output is A and B.
- Program B stores the numbers `65` and `66` in a `char` variable. A `char` keeps every character as its ASCII code, and 65 is the code of 'A' while 66 is the code of 'B'. So `cout` prints the letters A and B, not the numbers.
- To print 65 and 66 instead, the variable has to be an `int` (for example `int letter = 'A';`), because an `int` is printed as a number.
- `'\n'` moves the output to a new line, and `cin >> letter;` at the end waits for input so the console window stays open.

Program 2.2

Question

Take two integers as input from the user, apply arithmetic operations on them (+, -, *, /) and print them on screen.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int num1, num2;
    cout << "Enter first integer: ";
    cout << "Enter second integer: ";
    cin >> num1;
    cin >> num2;
    cout << "Addition: " << num1 << " + " << num2 << " = " << num1 + num2 << endl;
    cout << "Sub: " << num1 << " - " << num2 << " = " << num1 - num2 << endl;
    cout << "Multi: " << num1 << " * " << num2 << " = " << num1 * num2 << endl;
    cout << "Division: " << num1 << " / " << num2 << " = " << num1 / num2 << endl;
    return 0;
}
```

Output

```
Enter first integer: Enter second integer: 20 5
Addition: 20 + 5 = 25
Sub: 20 - 5 = 15
Multi: 20 * 5 = 100
Division: 20 / 5 = 4
```

Explanation

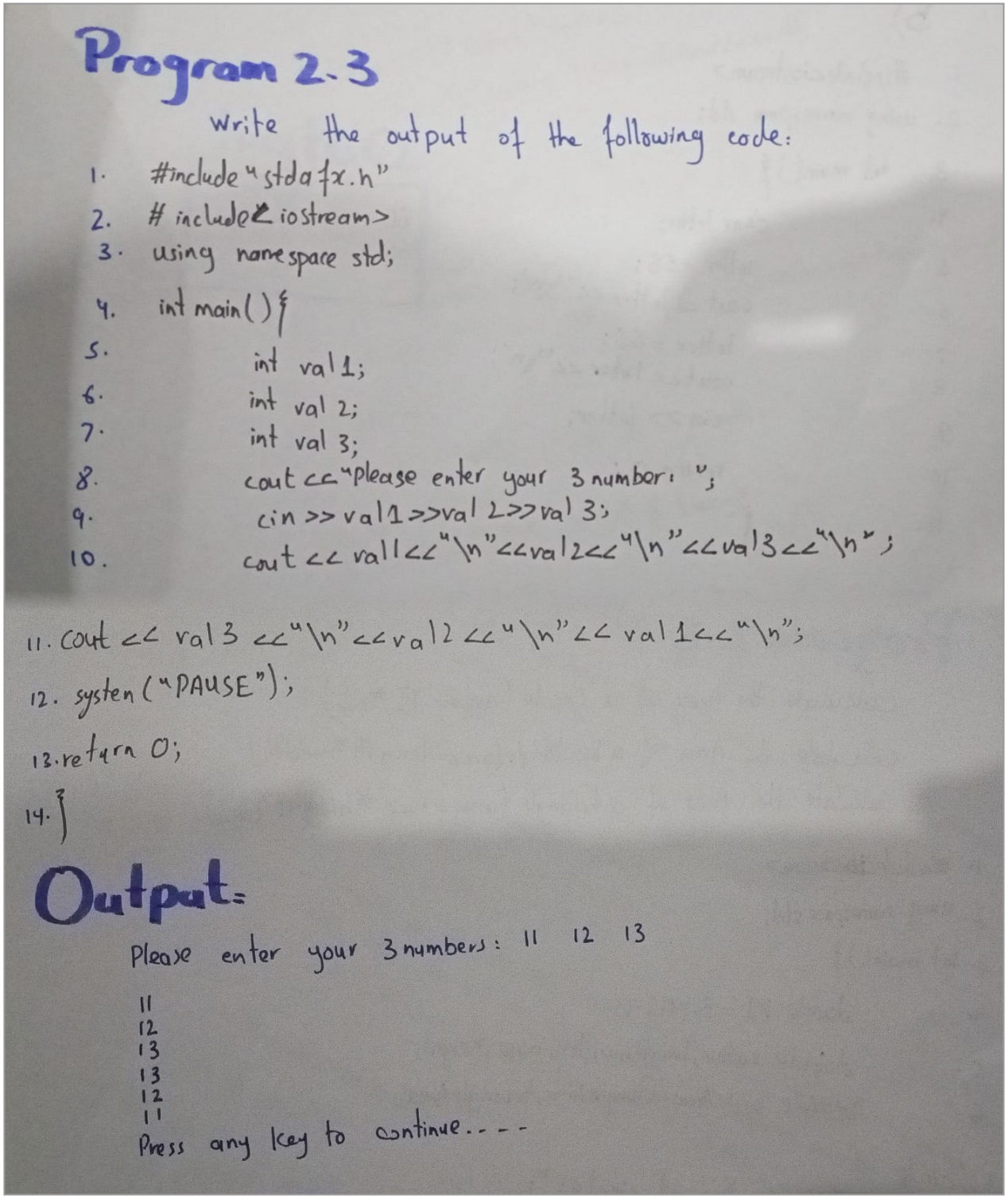
- `num1` and `num2` are `int` variables that store the two numbers typed by the user.
- Both prompts are printed first and then `cin` reads the two numbers, so the user types both numbers together (for example `20 5`).
- Each `cout` line prints the expression and its answer using the `+`, `-`, `*` and `/` operators.
- Because both numbers are integers, `/` does integer division and drops the decimal part. For example 7 / 2 gives 3, not 3.5.

Program 2.3

Question

Write the output of the following code.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include "stdafx.h"
#include <iostream>
using namespace std;
int main() {
    int val1;
    int val2;
    int val3;
    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;
    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
    system("PAUSE");
    return 0;
}
```

Output

```
Please enter your 3 numbers: 11 12 13
11
12
13
13
12
11
Press any key to continue . . .
```

Explanation

- `#include "stdafx.h"` is a Visual Studio precompiled header. Other compilers such as Dev-C++ do not need it, so it can be removed there.
- `cin >> val1 >> val2 >> val3;` reads all three numbers in one line.
- The first `cout` prints the numbers in the same order they were entered, and the second `cout` prints them in reverse order. `"\n"` puts each number on its own line.
- `system("PAUSE");` shows "Press any key to continue . . ." and waits, so the output can be read before the window closes.

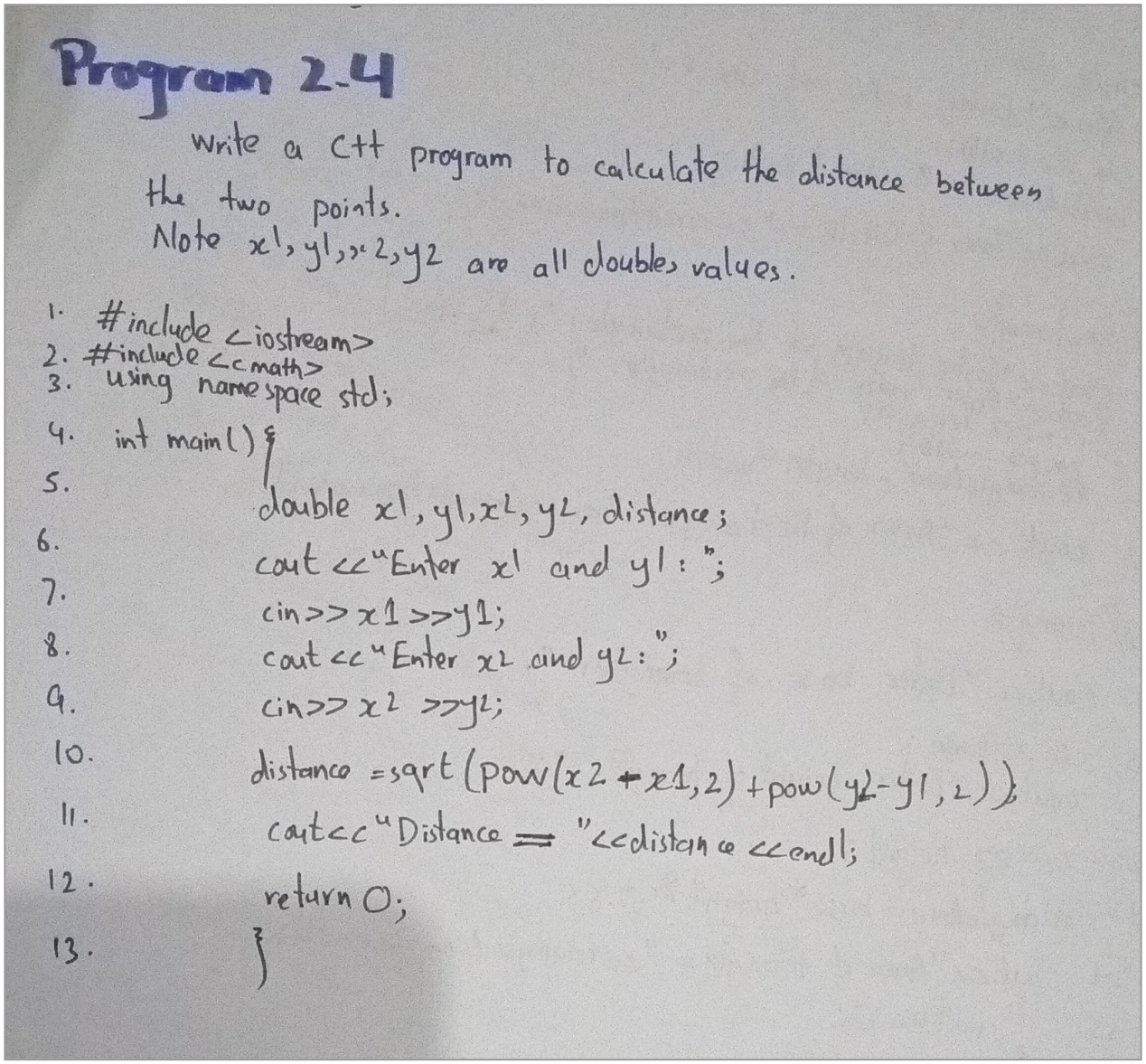
Program 2.4

Question

Write a C++ program to calculate the distance between the two points. Note: x1, y1, x2, y2 are all double values.

```
Formula:  d = √( (x2 − x1)² + (y2 − y1)² )
```

Handwritten Code



Program 2.4

Write a C++ program to calculate the distance between the two points.
Note x1, y1, x2, y2 are all doubles values.

```
1.  #include <iostream>
2.  #include <cmath>
3.  using namespace std;
4.  int main(){
5.      double x1, y1, x2, y2, distance;
6.      cout<<"Enter x1 and y1: ";
7.      cin>>x1>>y1;
8.      cout<<"Enter x2 and y2: ";
9.      cin>>x2>>y2;
10.     distance =sqrt(pow(x2-x1,2)+pow(y2-y1,2));
11.     cout<<"Distance = "<<distance<<endl;
12.     return 0;
13. }
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main() {
    double x1, y1, x2, y2, distance;
    cout << "Enter x1 and y1: ";
    cin >> x1 >> y1;
    cout << "Enter x2 and y2: ";
    cin >> x2 >> y2;
    distance = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
    cout << "Distance = " << distance << endl;
    return 0;
}
```

Output

```
Enter x1 and y1: 1 2
Enter x2 and y2: 4 6
Distance = 5
```

Explanation

- `#include <cmath>` gives us the math functions `sqrt()` (square root) and `pow()` (power).
- `pow(x2 - x1, 2)` squares the difference of the x values and `pow(y2 - y1, 2)` squares the difference of the y values.
- `sqrt()` of their sum gives the distance, exactly like the formula.
- Example: for the points (1, 2) and (4, 6) the distance is $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.

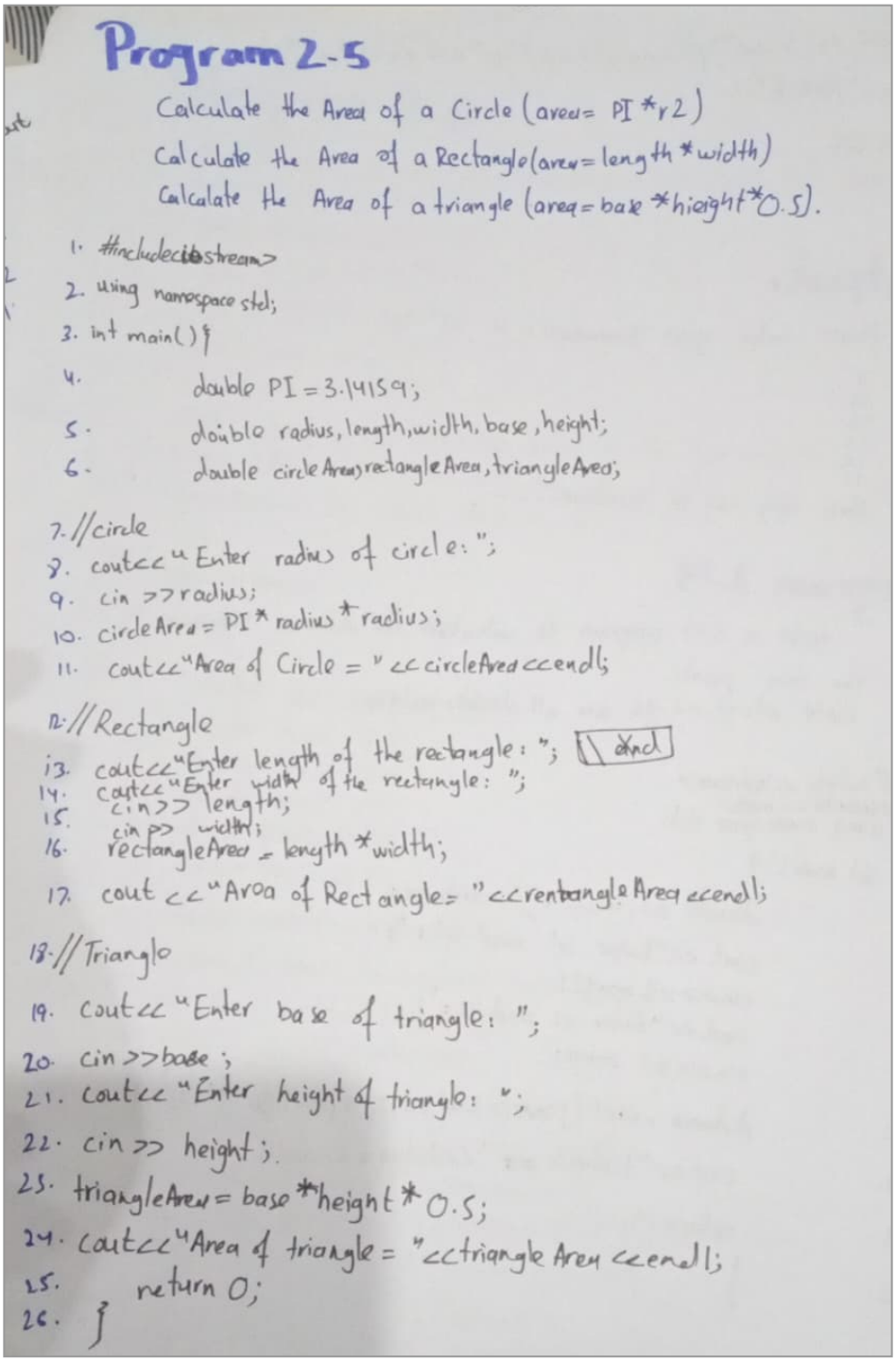
Program 2.5

Question

Calculate the area of a circle, a rectangle and a triangle:

1. `Circle     area = PI * r * r`
2. `Rectangle  area = length * width`
3. `Triangle   area = base * height * 0.5`

Handwritten Code



Program 2-5

Calculate the Area of a Circle (area= PI *r2)
Calculate the Area of a Rectangle(area=length*width)
Calculate the Area of a triangle (area=base*hieight*0.5).

```
1. #includeciostream>
2. using namespace std;
3. int main(){
4.         double PI=3.14159;
5.         double radius,length,width,base,height;
6.         double circleArea,rectangleArea,triangleArea;

7. //circle
8. coutcc"Enter radius of circle:";
9. cin >>radius;
10. circleArea = PI * radius * radius;
11. coutcc"Area of Circle = "cc circleAreaccendl;

12.//Rectangle
13. coutcc"Enter length of the rectangle:"; #incl
14. coutcc"Enter width of the rectangle:";
15.  cin>> length;
16.  cin>> width;
     rectangleArea = length *width;
17. cout cc"Area of Rectangle= "ccrentangleAreaccendl;

18.//Trianglo
19. coutcc"Enter base of triangle:";
20. cin >>base;
21. coutcc"Enter height of triangle:";
22. cin >> height;
23. triangleArea= base*height*0.5;
24. coutcc"Area of triangle = "cctriangle Areaccendl;
25.     return 0;
26. }
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    double PI = 3.14159;
    double radius, length, width, base, height;
    double circleArea, rectangleArea, triangleArea;

    // Circle
    cout << "Enter radius of circle: ";
    cin >> radius;
    circleArea = PI * radius * radius;
    cout << "Area of Circle = " << circleArea << endl;

    // Rectangle
    cout << "Enter length of the rectangle: ";
    cin >> length;
    cout << "Enter width of the rectangle: ";
    cin >> width;
    rectangleArea = length * width;
    cout << "Area of Rectangle = " << rectangleArea << endl;

    // Triangle
    cout << "Enter base of triangle: ";
    cin >> base;
    cout << "Enter height of triangle: ";
    cin >> height;
    triangleArea = base * height * 0.5;
    cout << "Area of Triangle = " << triangleArea << endl;
    return 0;
}
```

Output

```
Enter radius of circle: 5
Area of Circle = 78.5397
Enter length of the rectangle: 4
Enter width of the rectangle: 6
Area of Rectangle = 24
Enter base of triangle: 10
Enter height of triangle: 5
Area of Triangle = 25
```

Explanation

- `PI` stores 3.14159, and all values are `double` so decimal numbers also work.
- Circle: `PI * radius * radius` (πr^2). A radius of 5 gives 78.5397.
- Rectangle: `length * width`, so 4 × 6 = 24.
- Triangle: `base * height * 0.5`, so 10 × 5 × 0.5 = 25.
- Lines starting with `//` are comments. They label each part of the program and are ignored by the compiler.

Program 2.6

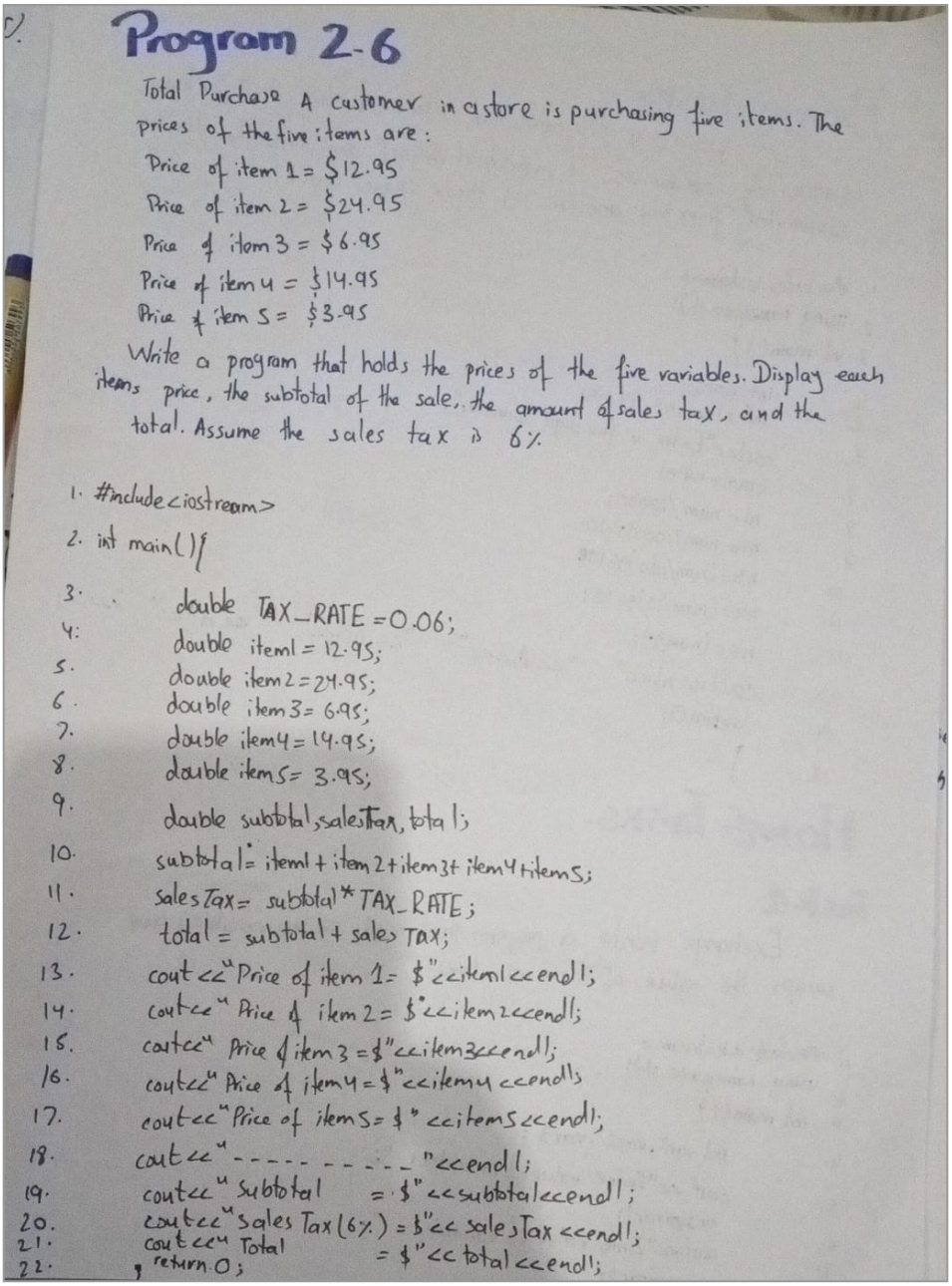
Question

Total Purchase: A customer in a store is purchasing five items. The prices of the five items are:

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
```

Write a program that holds the prices of the five items in five variables. Display each item's price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 6%.

Handwritten Code



Program 2-6

Total Purchase A customer in a store is purchasing five items. The prices of the five items are:

Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95

Write a program that holds the prices of the five variables. Display each items price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 6%.

```
1. #include<iostream>
2. int main(){
3.     double TAX_RATE=0.06;
4:     double item1 = 12.95;
5.     double item2=24.95;
6.     double item3= 6.95;
7.     double item4=14.95;
8.     double item5= 3.95;
9.     double subtotal,salesTax, total;
10.    subtotal= item1+item2+item3+item4+item5;
11.    salesTax= subtotal*TAX_RATE;
12.    total = subtotal+salesTax;
13.    cout<<"Price of item 1= $"<<item1<<endl;
14.    cout<<" Price of item 2= $"<<item2<<endl;
15.    cout<<" Price of item 3 =$"<<item3<<endl;
16.    cout<<" Price of item4 =$"<<item4<<endl;
17.    cout<<"Price of item5= $"<<item5<<endl;
18.    cout<<"----------"<<endl;
19.    cout<<" subtotal    = $"<<subtotal<<endl;
20.    cout<<"sales Tax(6%) = $"<<salesTax<<endl;
21.    cout<<" Total       = $"<<total<<endl;
22.    return 0;
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    double TAX_RATE = 0.06;
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;
    double subtotal, salesTax, total;
    subtotal = item1 + item2 + item3 + item4 + item5;
    salesTax = subtotal * TAX_RATE;
    total = subtotal + salesTax;
    cout << "Price of item 1 = $" << item1 << endl;
    cout << "Price of item 2 = $" << item2 << endl;
    cout << "Price of item 3 = $" << item3 << endl;
    cout << "Price of item 4 = $" << item4 << endl;
    cout << "Price of item 5 = $" << item5 << endl;
    cout << "-------------------------" << endl;
    cout << "Subtotal        = $" << subtotal << endl;
    cout << "Sales Tax (6%)  = $" << salesTax << endl;
    cout << "Total           = $" << total << endl;
    return 0;
}
```

Output

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
-------------------------
Subtotal        = $63.75
Sales Tax (6%)  = $3.825
Total           = $67.575
```

Explanation

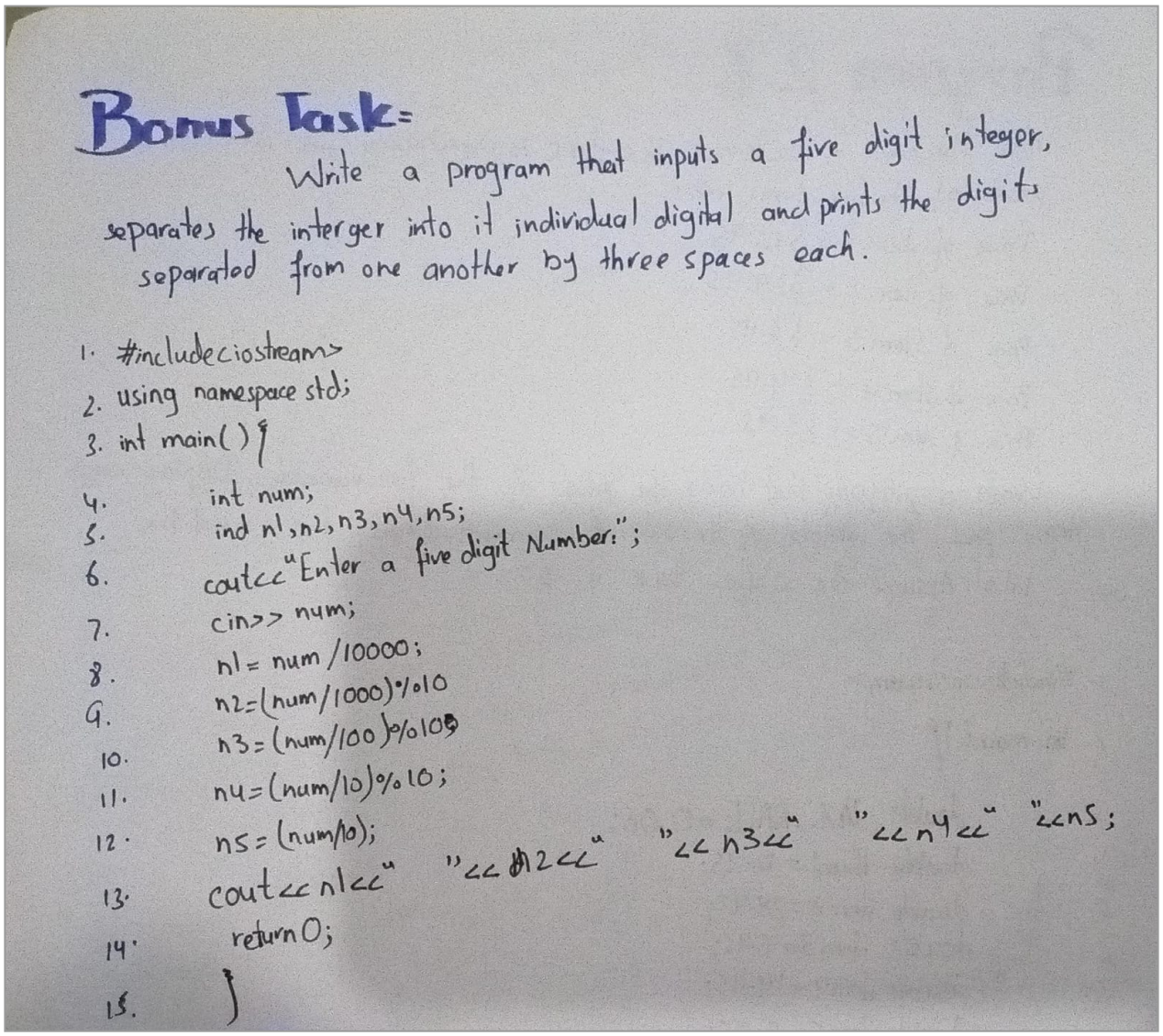
- The five prices are stored in five `double` variables, and `TAX_RATE` stores 6% as 0.06.
- `subtotal` adds the five prices together, which gives 63.75.
- `salesTax = subtotal * TAX_RATE;` gives 3.825, and `total = subtotal + salesTax;` gives 67.575.
- The dashed line only separates the item prices from the totals so the bill is easy to read.

Bonus Task

Question

Break Number: Write a program that inputs a five-digit integer, separates the integer into its individual digits and prints the digits separated from one another by three spaces each.

Handwritten Code



Bonus Task=

Write a program that inputs a five digit integer, separates the interger into it individual digital and prints the digits separated from one another by three spaces each.

```
1. #include<iostream>
2. using namespace std;
3. int main(){
4.       int num;
5.       ind n1,n2,n3,n4,n5;
6.       cout<<"Enter a five digit Number:";
7.       cin>> num;
8.       n1= num /10000;
9.       n2=(num/1000)%10
10.      n3=(num/100)%10
11.      n4=(num/10)%10;
12.      n5=(num/10);
13.      cout<<n1<<"  "<<n2<<"  "<<n3<<"  "<<n4<<"  "<<n5;
14.      return 0;
15.   }
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int num;
    int n1, n2, n3, n4, n5;
    cout << "Enter a five digit number: ";
    cin >> num;
    n1 = num / 10000;
    n2 = (num / 1000) % 10;
    n3 = (num / 100) % 10;
    n4 = (num / 10) % 10;
    n5 = num % 10;
    cout << n1 << "   " << n2 << "   " << n3 << "   " << n4 << "   " << n5 << endl;
    return 0;
}
```

Output

```
Enter a five digit number: 42339
4   2   3   3   9
```

Explanation

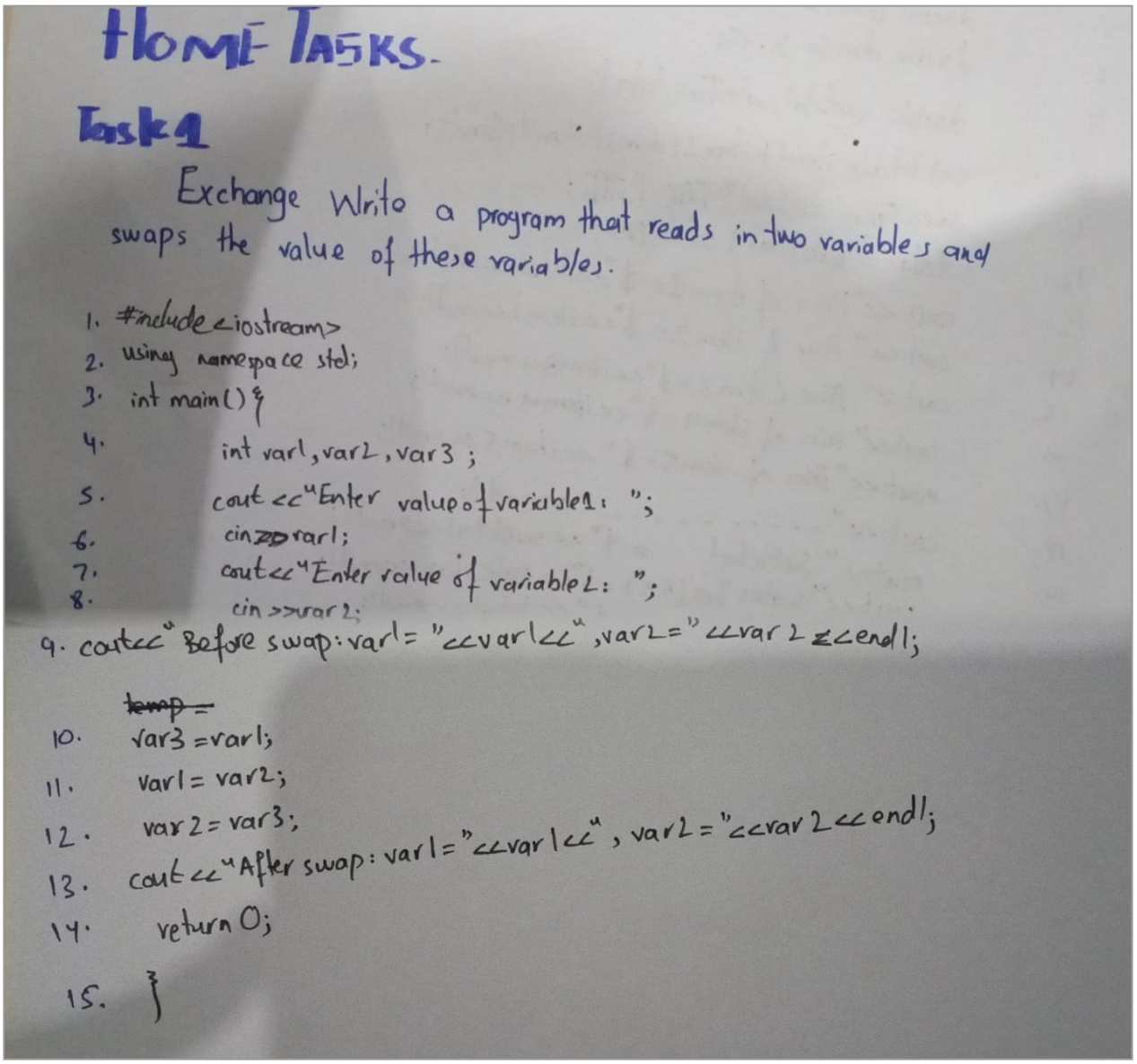
- On integers, `/` removes digits from the right and `% 10` gives the last digit. Using both together picks out one digit at a time.
- For 42339: `42339 / 10000` = 4, `(42339 / 1000) % 10` = 2, `(42339 / 100) % 10` = 3, `(42339 / 10) % 10` = 3 and `42339 % 10` = 9.
- `"   "` (three spaces) is printed between the digits, as the question asks.

Home Task 1

Question

Exchange: Write a program that reads in two variables and swaps the value of these variables. For example, if variable 'var1' contains 10 and variable 'var2' contains 20, after the swap operation 'var1' contains 20 and 'var2' contains 10.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int var1, var2, var3;
    cout << "Enter value of variable 1: ";
    cin >> var1;
    cout << "Enter value of variable 2: ";
    cin >> var2;
    cout << "Before swap: var1 = " << var1 << ", var2 = " << var2 << endl;
    var3 = var1;
    var1 = var2;
    var2 = var3;
    cout << "After swap: var1 = " << var1 << ", var2 = " << var2 << endl;
    return 0;
}
```

Output

```
Enter value of variable 1: 10
Enter value of variable 2: 20
Before swap: var1 = 10, var2 = 20
After swap: var1 = 20, var2 = 10
```

Explanation

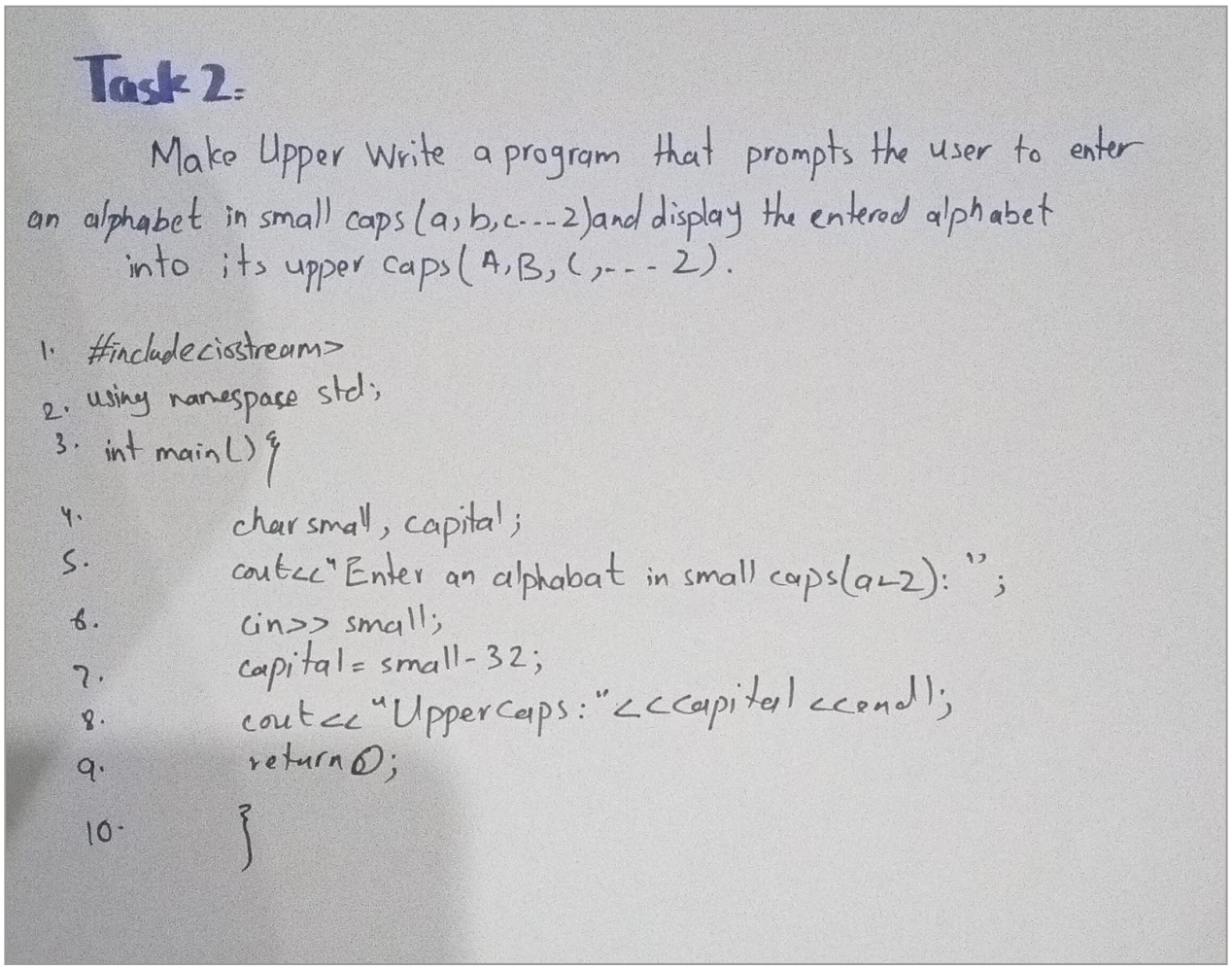
- `var3` is used as a temporary variable to hold a value during the swap.
- `var3 = var1;` saves the first value, `var1 = var2;` copies the second value into the first, and `var2 = var3;` puts the saved value into the second.
- Without the temporary variable, the first value would be lost as soon as `var1 = var2;` runs.
- The values are printed before and after the swap to show that they have been exchanged.

Home Task 2

Question

Make Upper: Write a program that prompts the user to enter an alphabet in small caps (a, b, c, ... z) and display the entered alphabet in its upper caps (A, B, C, ... Z). (Hint: ASCII)

Handwritten Code



Task 2:

Make Upper Write a program that prompts the user to enter an alphabet in small caps (a, b, c---z) and display the entered alphabet into its upper caps (A, B, C, ---Z).

```
1.  #include<iostream>
2.  using namespace std;
3.  int main() {
4.          char small, capital;
5.          cout<<"Enter an alphabat in small caps(a-z): ";
6.          cin>> small;
7.          capital= small-32;
8.          cout<<"Upper caps:"<<capital<<endl;
9.          return 0;
10.         }
```

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    char small, capital;
    cout << "Enter an alphabet in small caps (a-z): ";
    cin >> small;
    capital = small - 32;
    cout << "Upper Caps: " << capital << endl;
    return 0;
}
```

Output

```
Enter an alphabet in small caps (a-z): i
Upper Caps: I
```

Explanation

- A `char` is stored as its ASCII code. Small letters 'a' to 'z' are 97 to 122, and capital letters 'A' to 'Z' are 65 to 90.
- Every small letter is exactly 32 more than its capital letter. For example 'i' is 105 and 'I' is 73.
- So `capital = small - 32;` turns the small letter into its capital letter, and `cout` prints it as a character.

END OF ASSIGNMENT

```cpp
return 0;   // program finished successfully
```

Thank you for reading!